SOLUTION
SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (D)

Sol. About centre of circle :

$\omega_C = 2\omega = 0.8$ rad/sec

$v = \omega_C \cdot R = 0.4$ m/s

$a_C = \omega_C^2 R = 0.32$ m/sec

2. Ans. (A)

Sol. Time of travel from A to B, $t = \sqrt{\frac{2h}{g}}$ In this time, the particle must make an integral number of rotations

say n. $\Rightarrow 2\pi rn = ut \Rightarrow 2\pi rn = u\sqrt{\frac{2h}{g}} \Rightarrow n = \frac{u}{2\pi r}\sqrt{\frac{2h}{g}}$

3. Ans. (B)

Sol. According to string constant, point P is always pulled with an acceleration a along string which remains constant in magnitude.

4. Ans. (D)

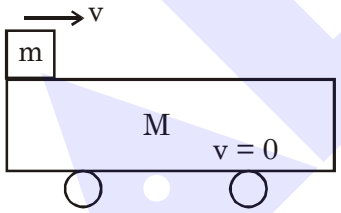
Sol. Suppose velocity of block at the end of slop = v

Then by work energy theorem

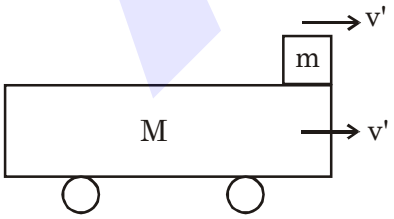
$$mgh = \frac{1}{2}mv^2$$

Now

Initially



Finally



Since there is no external force on (cart + block) system

$P_i = P_f$

$mv = (M + m)v'$

$$v' = \frac{mv}{M+m}$$

Now work done by friction = change in kinetic energy of system

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$$\therefore\ -\mu mg\ell = \frac{1}{2}(M+m)v'^2 - \frac{1}{2}mv^2$$

$$\mu mg\ell = \frac{1}{2}mv^2 - \frac{1}{2}(m+M)\left(\frac{m^2v^2}{(m+M)^2}\right)$$

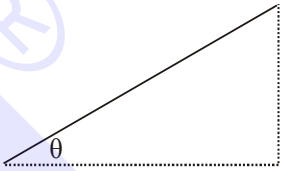
$$\mu mg\ell = \frac{1}{2}\left(\frac{mM}{m+M}\right)v^2;\ \mu mg\ell = \left(\frac{M}{m+M}\right)mgh;\ h = \mu\left(\frac{m+M}{M}\right)\ell$$

5. Ans. (C)

Sol. The speed at angle θ is given by conservation of energy

$\Rightarrow \frac{1}{2}mv^2 = mgh \Rightarrow \frac{1}{2}mv^2 = mg\,R\cos\theta \Rightarrow v = \sqrt{2gR\cos\theta}$ R will cancel out in the final answer.

So we have the following setup of projectile motion.



$v = \sqrt{2gR\cos\theta}$

$v_x = v\cos\theta$

$v_y = v\sin\theta$

The time of flight is twice the time to get to the top $\Rightarrow t = 2\left(\frac{v_y}{g}\right)$

$$\Rightarrow x = v_x t = v_x\left(\frac{2v_y}{g}\right) = \frac{2v_xv_y}{g} = \frac{2v_xv_y}{g} = \frac{2(v\cos\theta)(v\sin\theta)}{g} = \frac{2v^2\sin\theta\cos\theta}{g} = \frac{2(2gR\cos\theta)\sin\theta\cos\theta}{g}$$

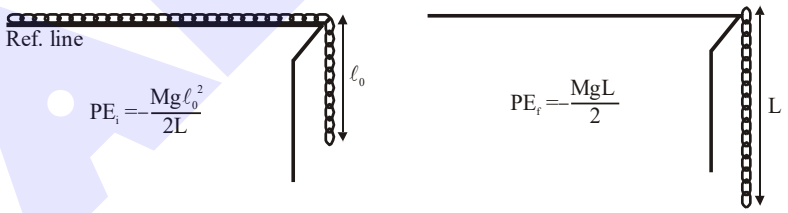
$$= 4R\cos^2\theta\sin\theta$$

Maximise this $\Rightarrow$ take the derivative and equate it to zero

$$\frac{dx}{d\theta} = -8R\cos\theta\sin^2\theta + 4R\cos^3\theta = 0 \Rightarrow 2\sin^2\theta = \cos^2\theta \Rightarrow \tan^2\theta = \frac{1}{2} \Rightarrow \tan\theta = \frac{1}{\sqrt{2}}$$

6. Ans. (C)

Sol. Potential energy of centre of mass is equal to potential energy of a system.



Loss in potential energy = gain in kinetic energy $-\frac{Mg}{2}\left[\frac{\ell_0^2}{L} - L\right] = \frac{1}{2}MV^2 \Rightarrow V = \sqrt{\frac{g}{L}\left(L^2 - \ell_0^2\right)}$

7. Ans. (B)

Sol. Loss in potential energy = gain in kinetic energy

$$mg\left(\frac{h}{\cos\theta} - h\right) = \frac{1}{2}mv^2$$

8. Ans. (D)

Sol. $\frac{dy}{dx} = \tan\theta = x$

tangential acceleration = gsinθ

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Multiple Correct Answer Type 3 Q. [4 M (–1)]

9. Ans. (A, B, C)

Sol. $\vec{r} = ut[\cos\omega t\hat{i} + \sin\omega t\hat{j}]$

$$\Rightarrow \vec{v} = \frac{d\vec{r}}{dt} = u[\cos\omega t\hat{i} + \sin\omega t\hat{j}] + ut[-\omega\sin\omega t\hat{i} + \omega\cos\omega t\hat{j}]$$

$$|a_1| = \frac{d|\vec{v}|}{dt} = \frac{\omega^2 tu}{\sqrt{1+\omega^2t^2}}$$

10. Ans. (A,B,C)

Sol. $\vec{F} = \frac{3\vec{r}}{r^3} = \frac{3\hat{r}}{r^2}$

$\vec{F}$ is a constant force and constant force is always conservative.

11. Ans. (ACD)

$$T = \frac{mv^2}{\ell}$$

Linked Comprehension Type (1 Para × 2Q, 1 Para × 3 Q.) [3 M(-1)]
(Single Correct Answer Type)

12. Ans. (D)

Sol. Force on block along slot = $m\omega^2 r = ma = m\left(\frac{vdv}{dr}\right)$

$$\int_0^v vdv = \int_{R/2}^r \omega^2 rdr$$

$$\frac{v^2}{2} = \frac{\omega^2}{2}\left(r^2 - \frac{R^2}{4}\right) \Rightarrow v = \omega\sqrt{r^2 - \frac{R^2}{4}} = \frac{dr}{dt}$$

$$\Rightarrow \int_{R/4}^r \frac{dr}{\sqrt{r^2 - \frac{R^2}{4}}} = \int_0^t \omega dt$$

$$\ell n\left(\frac{r + \sqrt{r^2 - \frac{R^2}{4}}}{\frac{R}{2}}\right) - \ell n\left(\frac{R/2 + \sqrt{\frac{R^2}{4} - \frac{R^2}{4}}}{\frac{R}{2}}\right) = \omega t$$

$$\Rightarrow r + \sqrt{r^2 - \frac{R^2}{4}} = \frac{R}{2}e^{\omega t}$$

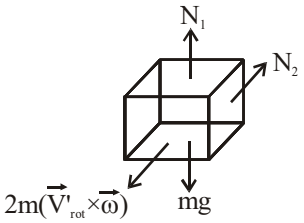
$$\Rightarrow r^2 - \frac{R^2}{4} = \frac{R^2}{4}e^{2\omega t} + r^2 - 2r\frac{R}{2}e^{\omega t}$$

$$\Rightarrow r = \frac{\frac{R^2}{4}e^{2\omega t} + \frac{R^2}{4}}{Re^{\omega t}} = \frac{R}{4}(e^{\omega t} + e^{-\omega t})$$

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13. Ans. (C)

Sol. $\vec{N}_1 = mg\hat{k}$



$$\vec{N}_2 = 2m\left(V'_{rot}\times\vec{\omega}\right)\hat{j} \quad = 2m\left[\frac{\omega R}{4}\left(e^{\omega t}-e^{-\omega t}\right)\right]\omega\hat{j}$$

$$= \frac{1}{2}m\omega^2 R\left(e^{\omega t}-e^{-\omega t}\right)\hat{j}$$

Total reaction on block = $\vec{N}_1 + \vec{N}_2$

$$= \frac{1}{2}m\omega^2 R\left(e^{\omega t}-e^{-\omega t}\right)\hat{j} + mg\hat{k}$$

14. Ans. (B)

Sol. by definition

$$U_r - U_\infty = -\int_\infty^r \left(\frac{-2\alpha}{r^3}+\frac{\beta}{r^2}\right)\cdot dr$$

$$= -\int_\alpha^r -\frac{2\alpha}{r^3}\cdot dr - \int_\alpha^r \frac{\beta}{r^2}\cdot dr$$

$$= +\frac{2\alpha}{(-2)r^2}\Big|_\infty^r + \beta\frac{1}{r}\Big|_\infty^r$$

$$\Rightarrow -\alpha\left(\frac{1}{r^2}-0\right)+\square\,\beta\left(\frac{1}{r}-0\right) \Rightarrow -\frac{\alpha}{r^2}+\frac{\beta}{r}\ ]$$

15. Ans. (C)

Sol. At equilibrium $\vec{F}=0 \Rightarrow -\frac{2\alpha}{r_0^3}+\frac{\beta}{r_0^2}=0 \Rightarrow r_0=\frac{2\alpha}{\beta}$

Ionization energy $E_0 = -\Delta U = -(U_\infty - U_{r_0})$

$$\Rightarrow \left[0-\left(-\frac{\alpha}{r_0^2}+\frac{\beta}{r_0}\right)\right]$$

$$E_0 = \frac{\alpha}{(2\alpha/\beta)^2}-\frac{\beta}{(2\alpha/\beta)} \Rightarrow \frac{\beta^2}{4\alpha}-\frac{\beta^2}{2\alpha}=\square\ \frac{\beta^2}{4\alpha}$$

$$= |E_0| = \frac{\beta^2}{4\alpha}$$

16. Ans. (B)

Sol. Work required $\Rightarrow U_2 - U_1$ = work required.

$$W = U_2 - U_1 = -\frac{2\alpha}{r_0^2}+\frac{\beta}{r_0} = \left[-\frac{2\alpha}{(r_0/\sqrt{2})^2}+\frac{\beta}{(r_0/\sqrt{2})^2}\right]-\left[-\frac{2\alpha}{(\sqrt{2}r_0)^2}+\frac{\beta}{(\sqrt{2}r_0)^2}\right]\ ]$$

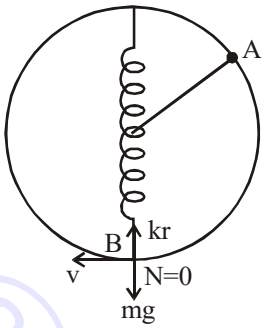
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SECTION-II

Numerical Answer Type Question **3 Q. [3(0)]**
(upto second decimal place)

17. **Ans.** 500 N/m

Sol. Initial length of spring = r (relaxed)



$$kr - mg = \frac{mv^2}{r} \qquad \text{... (i)}$$

COE at A & B

$$mgr\,(1+\cos 60°) - \frac{1}{2}kr^2 = \frac{1}{2}mv^2 \Rightarrow mv^2 = 3\,mgr$$

From equaiton (i)

$kr - mg = 3mg$

$kr = 4\,mg$

$$k = \frac{2mg}{r} = 500$$

18. **Ans. 40**

Sol. Let us denote the elastic constant (spring constant) of the rope by k and its unstretched length by ℓ_0. The maximum length of the rope is $\ell_1 = h - h_0 = 23m$, whilst in equilibrium it is $\ell_2 = (23 - 8)\ m = 15m$. Initially, and at the jumper's lowest position, the kinetic energy is zero. If we ignore the mass of the rope and assume that the jumper's centre of mass is half-way up his body, we can use conservation of energy to write.

$$mgh = \frac{1}{2}k(\ell_1 - \ell_0)^2.$$

In addition, in equilibrium,

$$mg = k\,(\ell_2 - \ell_0)$$

Dividing the two equations by each other we obtain a quadratic equation for ℓ_0.

$$\ell_0^2 + 2(h - \ell_1)\ell_0 + (\ell_1^2 - 2h\ell_2) = \ell_0^2 + 4\ell_0 - 221 = 0$$

which gives $\ell_0 = 13m$.

When the falling jumper attains his highest speed, his acceleration must be zero, and so this must occur at the same level as the final equilibrium position ($\ell = \ell_2$).

Again applying the law of conservation of energy,

$$\frac{1}{2}mv^2 + \frac{1}{2}k\,(\ell_2 - \ell_0)^2 = mg\,(\ell_2 + h_0)$$

where the ratio m/k is the same as that obtained from the equilibrium condition, namely,

$$\frac{m}{k} = \frac{\ell_2 - \ell_0}{g}$$

Substituting this into the energy equation, shows that the maximum speed of the jumper is $v = 18\ ms^{-1} \approx 65\ km\ h^{-1}$. It is easy to see that his maximum acceleration occurs at the lowest point of the jump. Since the largest extension of the rope (10 m) is five times that at the equilibrium position (2m), the greatest tension in the rope is 5mg. So the highest net force exerted on the jumper is 4mg, and his maximum acceleration is 4g.

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19. Ans. 729

Sol. $\frac{dv}{dt}=\frac{v^2}{R}\Rightarrow\int_{v_0}^{v}\frac{dv}{v^2}=\frac{1}{R}\int_0^t dt\Rightarrow\left(-\frac{1}{v}\right)_{v_0}^{v}=\frac{t}{R}\Rightarrow v=\frac{v_0}{1-\frac{v_0}{R}t}\Rightarrow\frac{ds}{dt}=\frac{v_0}{1-\frac{v_0}{R}t}\Rightarrow\int_0^{2\pi R}ds=\int_0^t\frac{v_0}{\left(1-\frac{v_0}{R}t\right)}dt$

$\Rightarrow 2\pi R=-R\left[\ell n\left(1-\frac{v_0}{R}t\right)\right]_0^t\Rightarrow 2\pi=-\ell n\left(1-\frac{v_0}{R}t\right)\Rightarrow 1-\frac{v_0}{R}t=e^{-2\pi}\Rightarrow t=\frac{R}{v_0}(1-e^{-2\pi})$

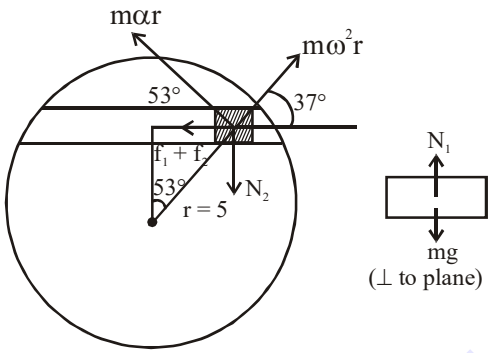
$\Rightarrow a=1,\beta=2\Rightarrow(\alpha+\beta)^6=(1+2)^6=729$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 8 Q. [4 M (0)]

20. Ans. 2

Sol. w.r.t table :-



$f_1=\mu N_1$; $f_2=\mu N_2$
also at time t
$\omega=\omega_0+\alpha t=t$

block start slide if resultant of $-\left(m\vec{a}_c+m\vec{a}_T\right)+\vec{N}_2$ become greater than maximum total friction

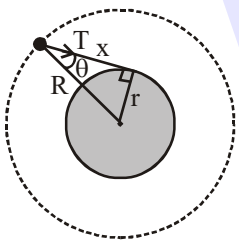
$\Rightarrow m\omega^2 r\cos 37°-m\alpha r\cos 53°>\mu(m\omega^2 r\sin 37°+m\alpha r\sin 53°)+\mu mg$

$\Rightarrow\frac{4t^2}{5}-\frac{3}{5}>\frac{1}{2}\left[\frac{3}{5}t^2+\frac{4}{5}+\frac{g}{r}\right]$

$\Rightarrow t^2>4\Rightarrow t>2$ sec.

21. Ans. 1

Sol.



$F_{CP}=ma_C=1\ (a_C)$
$F_C=T\cos\theta=a_C$
$a_C=2\ [x/R]$

$\because x=\sqrt{R^2-r^2}\Rightarrow\sqrt{\frac{4r^2}{3}-r^2}$ $\qquad$ **x =** $\frac{r}{\sqrt{3}}$

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$\therefore\ a_C = \dfrac{2\sqrt{R^2 - r^2}}{R}$

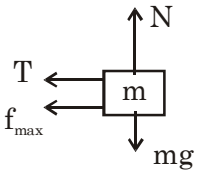
$a_C = 1\ m/s^2$

22. Ans. 7

Sol. As $vdv = \vec{a}.\vec{dr} = adx = 5dx \Rightarrow \int_0^v vdv = 5\int_0^{4.9} dx \Rightarrow \dfrac{v^2}{2} = 5(4.9) \Rightarrow v^2 = 49 \Rightarrow v = 7\ m/s$

23. Ans. 3

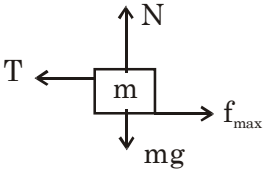
Sol. $T = mg$



For r_{max},

$T + f_{max} = m\omega^2 r_{max}$

$mg + \mu_S mg = m\omega^2 r_{max}$ .... (i)



For r_{min},

$T - f_{max} = m\omega^2 r_{min}$

$mg - \mu_S mg = m\omega^2 r_{min}$ .... (ii)

24. Ans. 3

Sol. (a) $m_1 g \sin\theta = (m_1 + m_1)a_1$

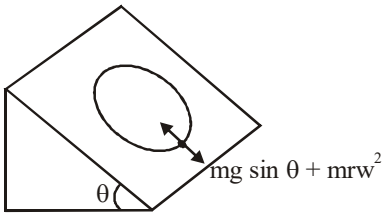
(b) $(m_2 - m_1)\, g \sin\theta/2 = (m_1 + m_2)\, a_2$

25. Ans. 6

Sol. $f \geq mg \sin\theta + mrw^2$

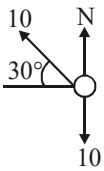
$\mu\, mg \cos\theta \geq mg \sin\theta + mrw^2$

or $\mu \geq \tan\theta + \dfrac{w^2 r}{g\cos\theta}$



26. Ans. 5

Sol. $T\cos\alpha = m\omega^2 (0.1)\cos\alpha$



$T = 1 \times 100 \times 0.1 = 10$

$10\sin 30° + N - 10 = 0 \Rightarrow N = 5$

27. Ans. 1

Sol. When the block has fallen by 60°, $\dfrac{1}{2}mv^2 = mgR(1 - \cos\theta) \Rightarrow v = \sqrt{2gR \times \dfrac{1}{2}} = \sqrt{gR}$

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Applying NLM along the radial direction $mg\cos\theta + kx = \frac{mv^2}{R} \Rightarrow x = \frac{1}{2}m$

Extension in spring = 0.5 m $\Rightarrow$Natural length = (R–x) = 1m

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

28. Ans. (A) Q, (B) P,Q,T (C) P,T (D) P,R,S

Sol. (A) Friction along tangent only.
(B) Friction provide centripetal and tangential acceleration.
(C) Friction provide only centripetal acceleration.
(D) If speed is not equal to $\sqrt{rg\tan\theta}$ then friction is along the incline of road to avoid slipping & if speed is equal to $\sqrt{rg\tan\theta}$, no friction.

Subjective Type **2 Q. [4 M (0)]**

29. Ans. (a) 2μg, (b) μmg/k, (c) $2\mu g\sqrt{\frac{m}{k}}$

Sol. (a) $ma = 3\mu mg - \mu mg \Rightarrow a = 2\mu g$
(b) From work-energy principle

$$\frac{1}{2}k(x_0^2 - x_1^2) - \mu mg(x_1 + x_0) = 0$$

$\Rightarrow \quad x_0 - x_1 = 2\mu mg/k \Rightarrow x_1 = \mu mg/k$

(c) Speed will be maximum where net force is zero
$\mu mg = kx \Rightarrow x = \mu mg/k$ (extension)
Now from work-energy principle

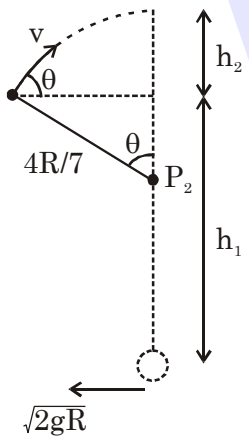
$$\frac{1}{2}k\left(\frac{\mu mg}{k}\right)^2 (3^2 - 1^2) - \frac{2\mu mg}{k}(\mu mg) = \frac{1}{2}mv^2$$

Solving we get

$$v = 2\mu g\sqrt{\frac{m}{k}}$$

30. Ans. 27R/28

Sol.



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At lowest point velocity of bob = $\sqrt{2gR}$

Angle where the string slack & bob start performing projecitle motion = θ

$$mg\cos\theta = \frac{mv^2}{\frac{4R}{7}} \qquad \text{... (i)}$$

Applying work energy theorem

$$-mg\left(\frac{4R}{7}\right)(1+\cos\theta) = \frac{1}{2}mv^2 - \frac{1}{2}m\left(\sqrt{2gR}\right)^2 \quad \text{.... (ii)}$$

Solving (i) & (ii)

$$\cos\theta = \frac{1}{2} \text{ and } V = \sqrt{\frac{2gR}{7}}$$

$$h_1 = \frac{4R}{7}(1+\cos\theta)$$

$$h_2 = \frac{v^2\sin^2\theta}{2g}$$

Maximum height from lowest point = $h_1 + h_2$

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